

3RD - WEB - HACK · TECHZAP CLUB

Eternal

Your keys die with you.

Your crypto doesn't have to.

non-custodial on-chain inheritance · Ethereum / any EVM · MIT

THE PROBLEM

\$140B+

of crypto sits in wallets whose owners are dead or unreachable. Self-custody has no "forgot password", no executor, no inheritance path.

exchanges = custodial

paper wills leak keys

multisig needs fluent heirs

existing tools charge rent

THE SOLUTION

A dead man's switch that can be **contested**

Seal

owner funds vault
heirs + shares
set



Heartbeat

every owner
action
proves liveness



Lapse

silence lets
heirs
file a claim



Challenge

owner cancels ·
guardian can
veto



Inherit

heirs pull
shares
+ letters + NFTs

no token · no oracle · no admin key · no subscription – one auditable contract

WHAT'S NEW

- 01 Sealed heirs** — heirs are keccak commitments, not addresses. Nobody can see or target them. A physical claim card is the only key.
- 02 Guardian veto** — an optional watchdog can cancel a premature claim. They can never touch funds; the contract forbids it.
- 03 Every action is a heartbeat** — depositing, editing heirs, sealing a letter all reset the timer. No ritual check-ins.
- 04 Specific NFT bequests** — "my Azuki goes to Alice", enforced on-chain, independent of percentage splits.
- 05 Sealed letters** — AES-256 encrypted in the browser, ciphertext on-chain, passphrase whispered off-chain. Last words survive.
- 06 Pull-based payouts** — heirs withdraw their own shares; shares are computed on frozen balances, so late claims can't dilute early ones.

BUILT & PROVEN

Not a concept. A running system.

- ▶ Solidity 0.8.24 – single contract, zero deps
- ▶ checks-effects-interactions · custom errors · events
- ▶ React + ethers v6 frontend – owner / heir / guardian portals
- ▶ AES-256-GCM letters via WebCrypto – client-side only

- ✓ 14 contract tests – full lifecycle + edge cases
- ✓ 21 headless-browser E2E checks – every button clicked
- ✓ full demo flow recorded end-to-end
- ✓ MIT · public repo · one-command local setup

IMPACT & FUTURE SCOPE

Who it's for

Every long-term self-custody holder is a single accident away from burning their family's savings. Eternal turns "telling your heirs the seed phrase" — today's only option, and a terrible one — into an enforceable on-chain will. Same primitive extends to DAO treasuries, founder key succession, and multisig disaster recovery.

NOW — HACKATHON

full lifecycle on EVM, sealed heirs, guardian veto, NFT bequests, letters

NEXT

Sepolia + L2 deploys · claim-card PDF export · email/Telegram lapse alerts

LATER

ERC-4337 session heartbeats · fiat-abstracted gas · estate bundles (multi-vault)

ETHERNAL

Seal → heartbeat → lapse →
challenge → inherit

MIT licensed · single auditable contract · live demo + full source in the repo
demo video: [demo/out/demo.mp4](#) · built for 3rd-Web-Hack